

Batch: A1 Roll No.: 16010123012**Experiment No.: 04****Grade: AA / AB / BB / BC / CC / CD / DD****Title:** Implementing Platform as a Service - Creating a Web app and deploying it on Azure Cloud**Objective:**

To Deploy a Python Web App on Microsoft Azure Platform and access the application on cloud through Browser

Expected Outcome of Experiment:

CO	Outcome
CO4	Build cloud services and applications

Books/ Journals/ Websites referred:

<https://learn.microsoft.com/en-us/azure/app-service/quickstart-python?tabs=flask%2Cwindows%2Cazure-cli%2Cazure-cli-deploy%2Cdeploy-instructions-azportal%2Cterminal-bash%2Cdeploy-instructions-zip-azcli>

Related Theory: -

Infrastructure as a Service (IaaS) is a cloud computing service model in which virtualized computing resources such as servers, storage, networking, and operating systems are provided over the internet on a pay-as-you-go basis. The cloud service provider manages the physical infrastructure, while the consumer has control over:

- Virtual machines
- Operating systems
- Storage
- Applications
- Networking configuration (to a large extent)

Key Components of IaaS are as under-

- Compute – Virtual Machines (VMs)
- Storage – Block, Object, File storage
- Networking – Virtual networks, subnets, IP addresses
- Virtualization Layer – Hypervisor abstracts physical hardware
- Management Interface – Web portal, CLI, APIs

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Responsibilities: Provider vs Consumer

Layer	Managed By
Physical Data Center	Cloud Provider
Servers & Storage Hardware	Cloud Provider
Virtualization (Hypervisor)	Cloud Provider
Operating System	Consumer
Middleware	Consumer
Runtime & Applications	Consumer
Data	Consumer

Characteristics of Infrastructure as a Service (IaaS)
1. On-demand self-service

Users can provision computing resources such as virtual machines and storage automatically without human interaction with the cloud provider. Resources can be created, modified, or deleted anytime using a web portal or API.

2. Scalability & Elasticity

IaaS allows resources to be scaled up or down based on workload requirements. This ensures optimal performance during peak demand and cost savings during low usage.

3. Resource Pooling

The cloud provider pools physical resources and dynamically allocates them to multiple users using virtualization. Customers share the same infrastructure while remaining logically isolated from each other.

4. Measured Service

Resource usage is continuously monitored, measured, and billed based on actual consumption. Users pay only for the compute, storage, and network resources they use.

5. High Availability

IaaS platforms provide reliable access to resources through redundant data centers and fault-tolerant architecture. This ensures minimal downtime and continuous service availability.

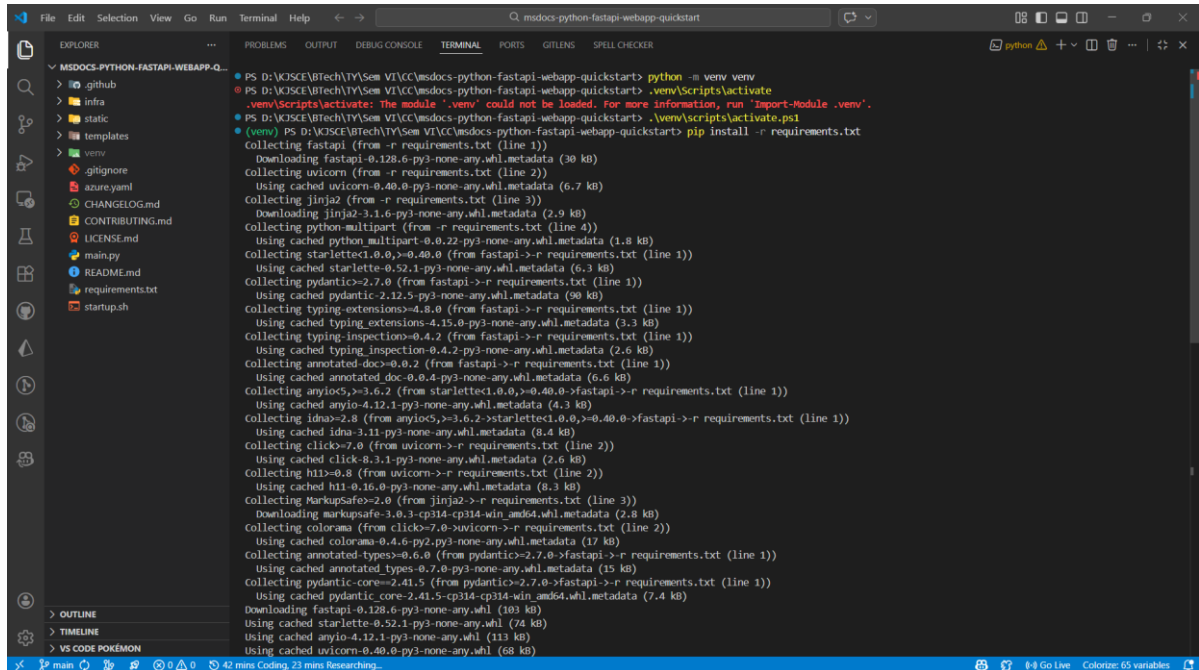
Implementation Details:

- Follow steps on available on this link:
<http://surl.li/pkbbv>
- Create a sample application (a web app) on a local machine using git cloning.

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3. Create a web service on Azure for hosting web app made in step 1. (USE AZURE PORTAL, NOT AZURE CLI)
4. Deploy code for application made in Step 1 on Azure cloud (DEPLOY USING LOCAL GIT)
5. Access app deployed on cloud through Browser

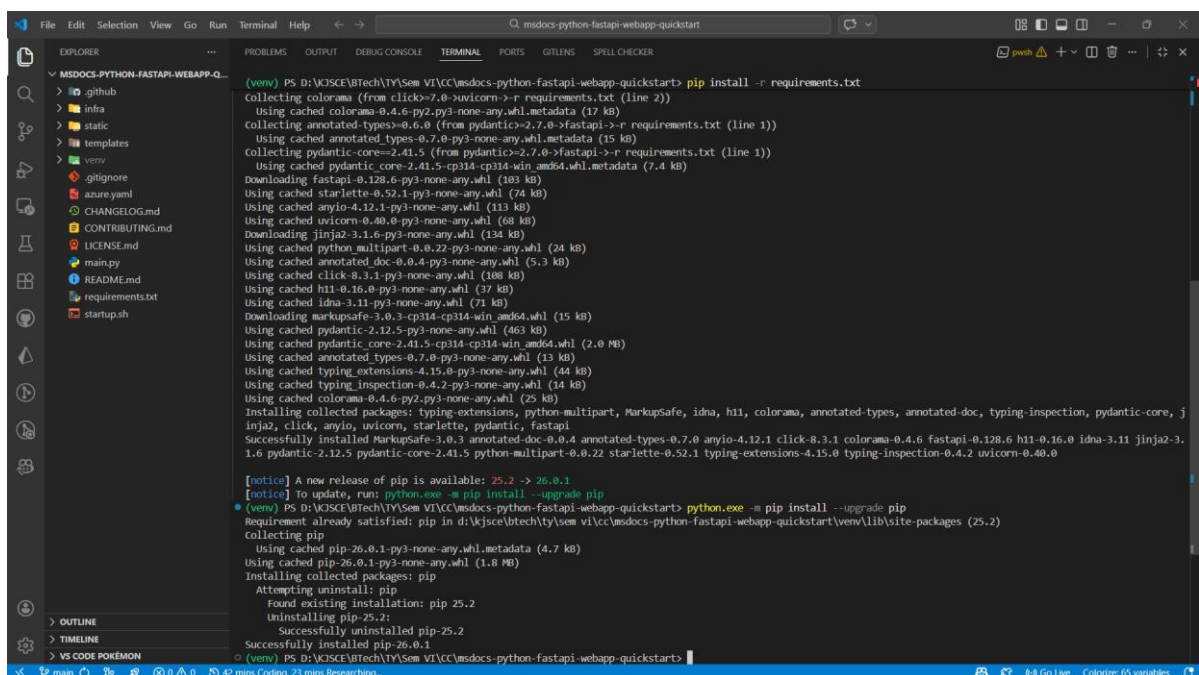
A. Include step wise screenshots in the document with proper heading for each screenshot.



```

PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> python -m venv venv
PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> .venv\Scripts\activate
.venv\Scripts\activate: The module '.venv' could not be loaded. For more information, run 'Import-Module .venv'.
PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> .venv\Scripts\activate.ps1
(venv) PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> pip install -r requirements.txt
Collecting fastapi (from -r requirements.txt (line 1))
  Downloading fastapi-0.128.6-py3-none-any.whl.metadata (30 kB)
Collecting uvicorn (from -r requirements.txt (line 2))
  Using cached uvicorn-0.40.0-py3-none-any.whl.metadata (6.7 kB)
Collecting jinja2 (from -r requirements.txt (line 3))
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Collecting pydantic-2.7.0 (from fastapi->-r requirements.txt (line 1))
  Using cached pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting typing-extensions-4.8.0 (from fastapi->-r requirements.txt (line 1))
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection-0.4.2 (from fastapi->-r requirements.txt (line 1))
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc-0.0.2 (from fastapi->-r requirements.txt (line 1))
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
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  Using cached anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting idna-3.11-py3-none-any.whl.metadata (8.4 kB)
  Using cached idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Collecting click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
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Collecting h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
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Collecting MarkupSafe-2.0 (from jinja2->-r requirements.txt (line 3))
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Collecting annotated-types-0.6.0 (from pydantic-2.7.0->-r requirements.txt (line 1))
  Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
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  Using cached pydantic_core-2.41.5-cp314-cp314-win_amd64.whl.metadata (7.4 kB)
  Downloading fastapi-0.128.6-py3-none-any.whl (103 kB)
  Using cached starlette-0.52.1-py3-none-any.whl (74 kB)
  Using cached anyio-4.12.1-py3-none-any.whl (113 kB)
  Using cached uvicorn-0.40.0-py3-none-any.whl (68 kB)
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  Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
  Using cached click-8.3.1-py3-none-any.whl (108 kB)
  Using cached idna-3.11-py3-none-any.whl (71 kB)
  Downloading MarkupSafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
  Using cached pydantic-2.12.5-py3-none-any.whl (463 kB)
  Using cached pydantic_core-2.41.5-cp314-cp314-win_amd64.whl (2.0 MB)
  Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)
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  Using cached typing_inspection-0.4.2-py3-none-any.whl (14 kB)
  Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: typing-extensions, python-multipart, MarkupSafe, idna, h11, colorama, annotated-types, annotated-doc, typing-inspection, pydantic-core, jinja2, click, anyio, uvicorn, starlette, pydantic, fastapi
Successfully installed MarkupSafe-3.0.3 annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 colorama-0.4.6 fastapi-0.128.6 h11-0.16.0 idna-3.11 jinja2-3.1.6 pydantic-2.12.5 pydantic-core-2.41.5 python-multipart-0.0.22 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.40.0
[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> python.exe -m pip install --upgrade pip
Requirement already satisfied: pip in d:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart\venv\lib\site-packages (25.2)
Collecting pip
  Using cached pip-26.0.1-py3-none-any.whl.metadata (4.7 kB)
  Using cached pip-26.0.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.2
    Uninstalling pip-25.2:
      Successfully uninstalled pip-25.2
  Successfully installed pip-26.0.1
(venv) PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart>
  
```

Cloning the Sample Application

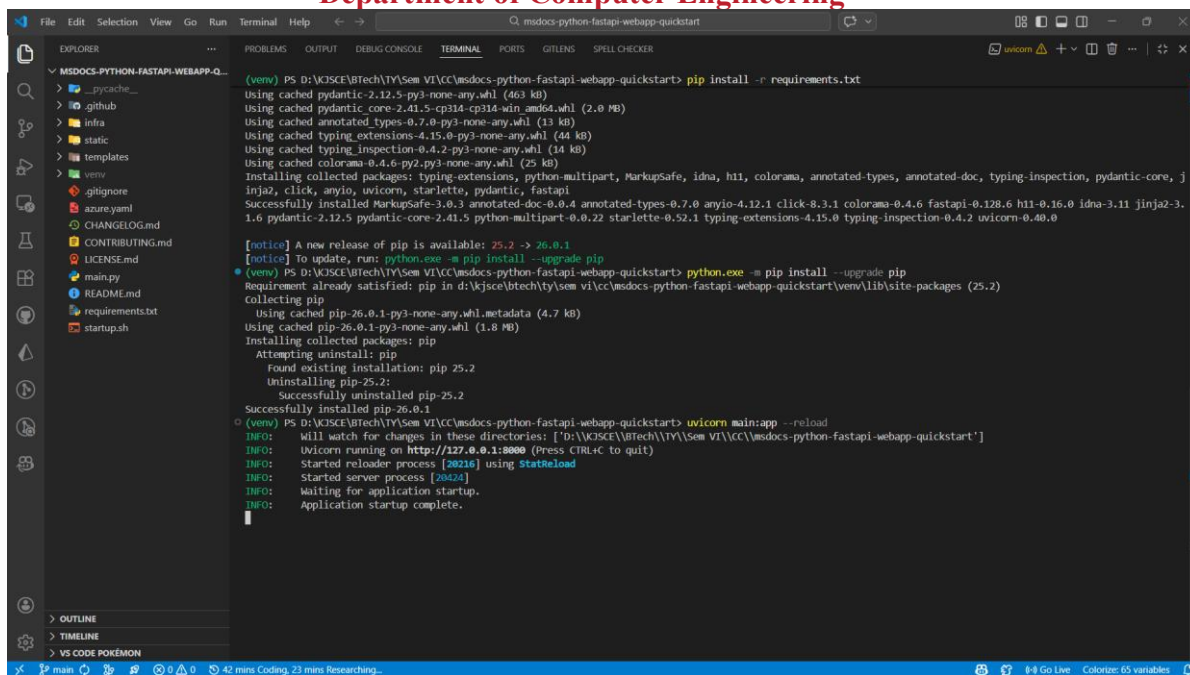


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(venv) PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> pip install -r requirements.txt
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Successfully installed MarkupSafe-3.0.3 annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 colorama-0.4.6 fastapi-0.128.6 h11-0.16.0 idna-3.11 jinja2-3.1.6 pydantic-2.12.5 pydantic-core-2.41.5 python-multipart-0.0.22 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.40.0
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  Successfully installed pip-26.0.1
(venv) PS D:\VCSE\BTEch\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart>
  
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Installing Dependencies

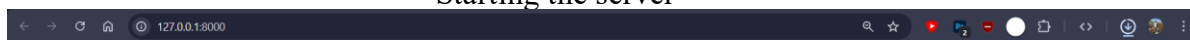
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(venv) PS D:\KJSCE\BTech\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> pip install -r requirements.txt
Using cached pydantic-2.12.5-py3-none-any.whl (463 kB)
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  Attempting uninstall: pip
    Found existing installation: pip 25.2
    Uninstalling pip-25.2:
      Successfully uninstalled pip-25.2
  Successfully installed pip-26.0.1
(venv) PS D:\KJSCE\BTech\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['D:\KJSCE\BTech\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [20216] using StatReload
INFO: Started server process [20224]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Starting the server



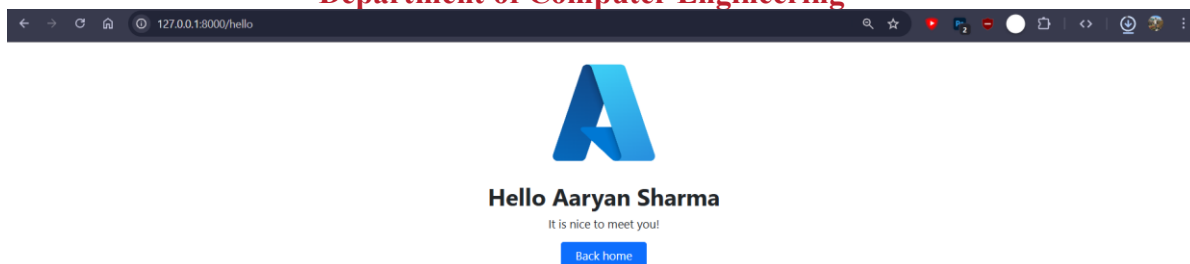
Welcome to Azure

Could you please tell me your name?

Aaryan Sharma

Say Hello

Browsing

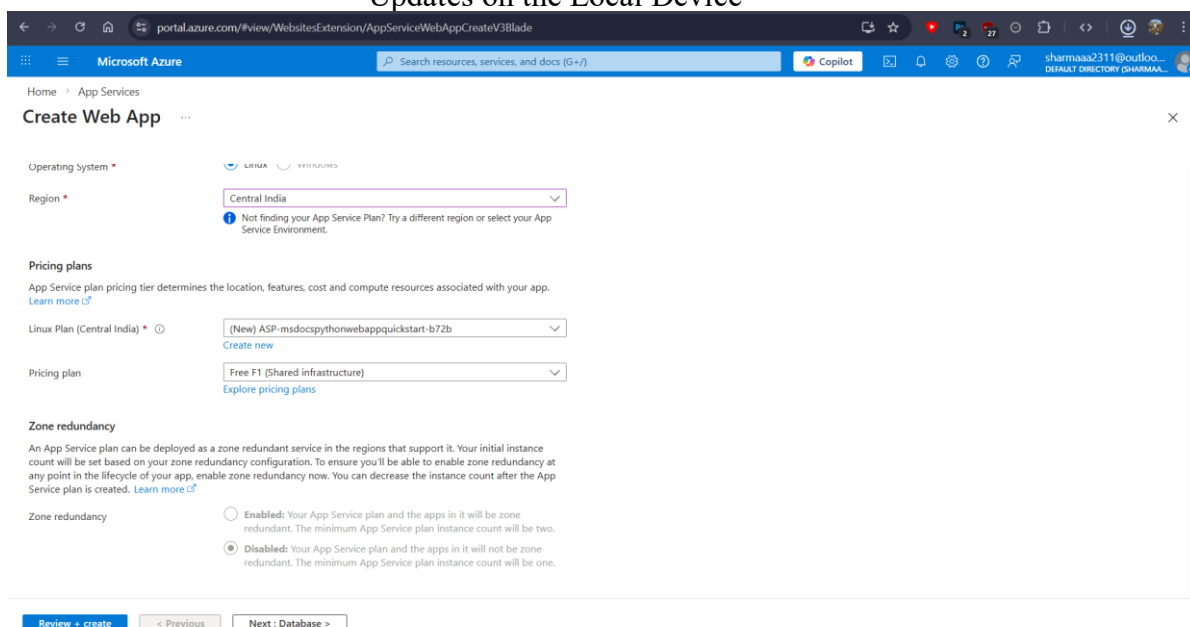


Browsing on LocalHost

```

(venv) PS D:\KJSCE\BTech\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['D:\KJSCE\BTech\TY\Sem VI\CC\msdocs-python-fastapi-webapp-quickstart']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [20216] using StatReload
INFO: Started server process [20424]
INFO: Waiting for application startup.
INFO: Application startup complete.
Request for index page received
INFO: 127.0.0.1:61915 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:59655 - "GET /static/images/azure-icon.svg HTTP/1.1" 200 OK
INFO: 127.0.0.1:61915 - "GET /static/bootstrap/css/bootstrap.min.css HTTP/1.1" 200 OK
INFO: 127.0.0.1:61915 - "GET /static/favicon.ico HTTP/1.1" 200 OK
Request for hello page received with name=Aaryan Sharma
INFO: 127.0.0.1:63948 - "POST /hello HTTP/1.1" 200 OK
INFO: 127.0.0.1:63948 - "GET /static/bootstrap/css/bootstrap.min.css HTTP/1.1" 304 Not Modified
INFO: 127.0.0.1:65126 - "GET /static/images/azure-icon.svg HTTP/1.1" 304 Not Modified
Request for index page received
INFO: 127.0.0.1:55255 - "GET / HTTP/1.1" 200 OK
  
```

Updates on the Local Device



Creating the Web App

portal.azure.com/#view/WebsitesExtension/AppServiceWebAppCreateV3Blade

Microsoft Azure Search resources, services, and docs (G+)

Home > App Services

Create Web App

Subscription: fd1b8626-e69e-4fb0-bec0-bff615360c49
Resource Group: msdocs-python-webapp-quickstart
Name: msdocs-python-webapp-quickstart-aar
Secure unique default hostname: Enabled
Publish: Code
Runtime stack: Python 3.14

App Service Plan (New)

Name: ASP-msdocspythonwebappquickstart-b72b
Operating System: Linux
Region: Central India
SKU: Free
ACU: Shared infrastructure
Memory: 1 GB memory

Monitor + secure

Application Insights: Not enabled

Deployment

Basic authentication: Disabled
Continuous deployment: Not enabled / Set up after app creation

[Create](#) [< Previous](#) [Next >](#) [Download a template for automation](#)

Review of the Web App before Creation

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~:/overview/id/%2Fsubscriptions%2Ffd1b8626-e69e-4fb0-bec0-bff615360c49...

Microsoft Azure Search resources, services, and docs (G+)

Home

Microsoft.Web-WebApp-Portal-6130d2b1-9e10 | Overview

Deployment

Search x < > Delete Cancel Redeploy Download Refresh

Overview

Inputs
Outputs
Template

Deployment is in progress

Deployment name : Microsoft.Web-WebApp-Portal-6130d2b1-9e10 Start time : 10/2/2026, 2:57:05 pm
Subscription : Azure for Students Correlation ID : 42f6e230-37d5-45d5-8668-0e35023f9c42
Resource group : msdocs-python-webapp-quickstart

Deployment details

Resource	Type	Status	Operation details
ASP-msdocspythonwebappquickstart-b72b	Microsoft.Web/serverfarms	OK	Operation details

Add or remove favorites by pressing Ctrl+Shift+F

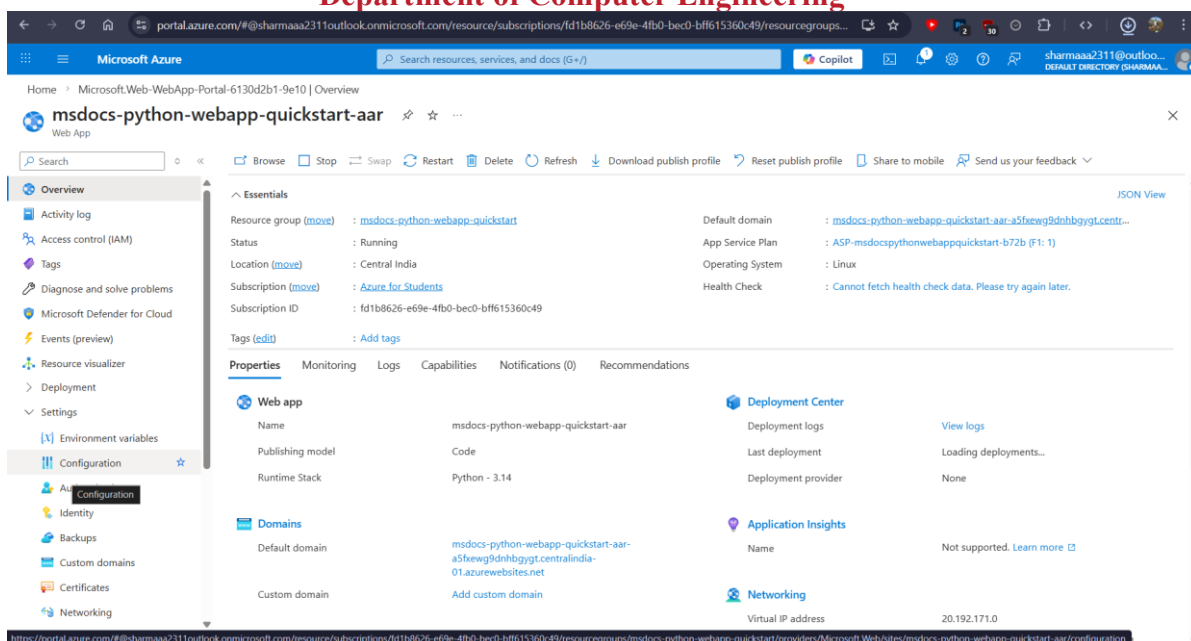
Microsoft Defender for Cloud
Secure your apps and infrastructure
[Go to Microsoft Defender for Cloud >](#)

Free Microsoft tutorials
[Start learning today >](#)

Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
[Find an Azure expert >](#)

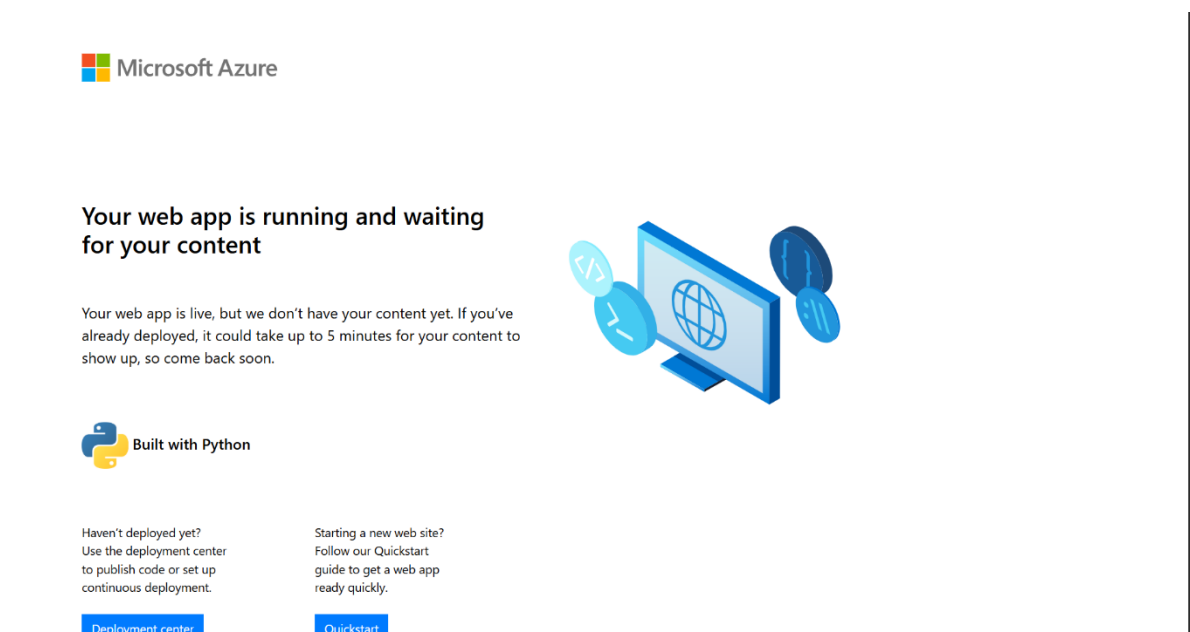
Web App Deployment

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The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Resource visualizer, Deployment, Settings, Environment variables, Configuration (selected), Auto Configuration, Identity, Backups, Custom domains, Certificates, and Networking. The main content area displays the 'Essentials' and 'Properties' tabs for the Web App. The 'Properties' tab shows details such as Resource group, Status (Running), Location (Central India), Subscription ID, and Tags. The 'Configuration' tab is active, showing the 'Web app' section with details like Name, Publishing model (Code), Runtime Stack (Python - 3.14), and Domains (Default domain and Custom domain). The 'Deployment Center' section shows deployment logs and the 'Application Insights' section indicates that it is not supported for this configuration.

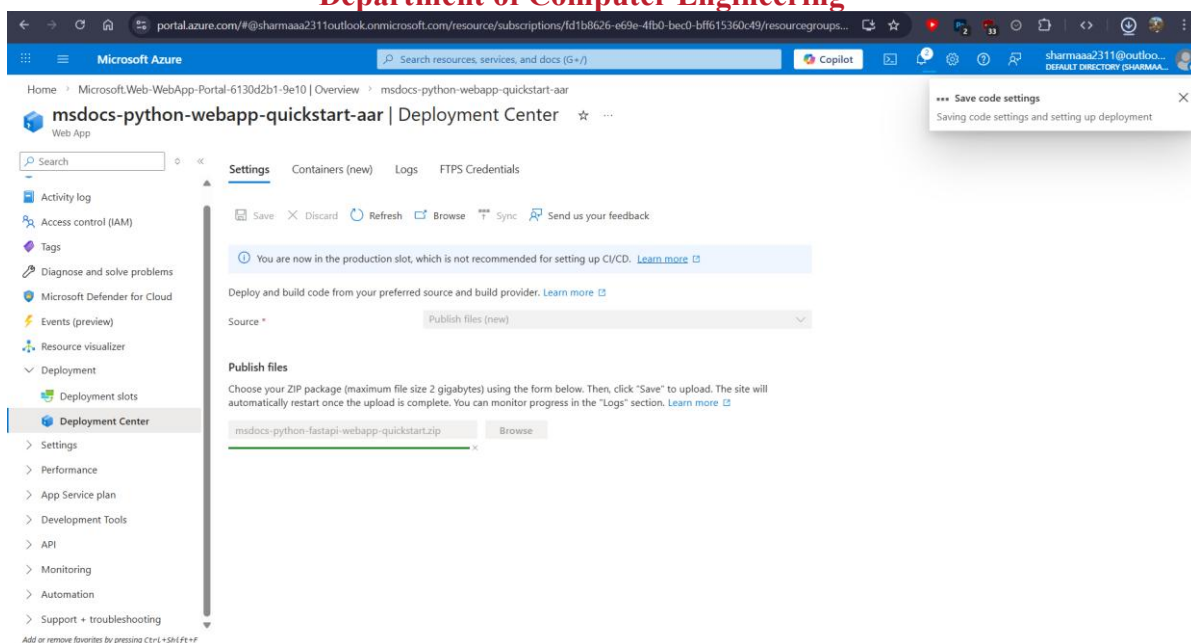
Opening the domain



The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Resource visualizer, Deployment, Settings, Environment variables, Configuration (selected), Auto Configuration, Identity, Backups, Custom domains, Certificates, and Networking. The main content area displays the 'Your web app is running and waiting for your content' message. The message states: 'Your web app is live, but we don't have your content yet. If you've already deployed, it could take up to 5 minutes for your content to show up, so come back soon.' Below the message, there are two buttons: 'Deployment center' and 'Quickstart'. The 'Quickstart' button is highlighted.

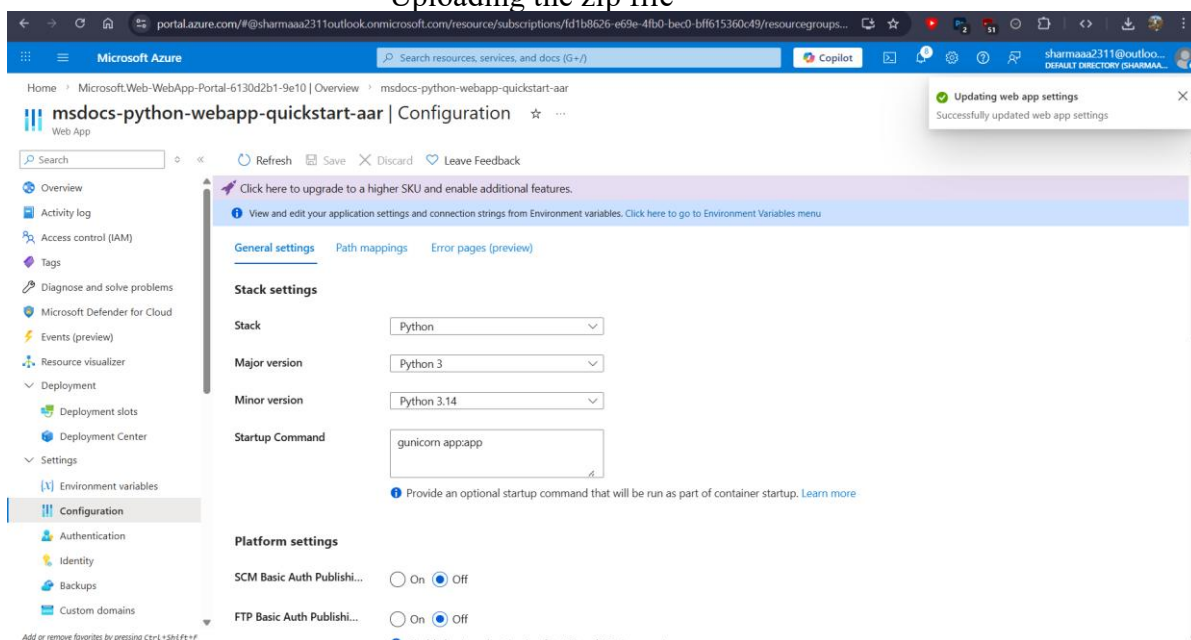
Domain after deployment

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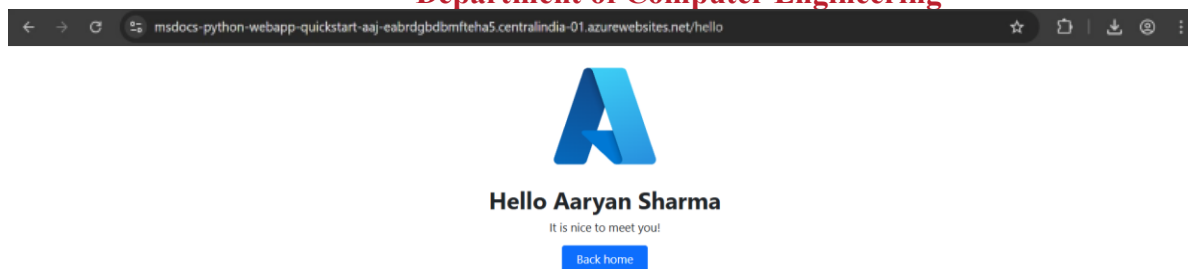
Microsoft Azure portal showing the **msdocs-python-webapp-quickstart-aar** Web App in the **Deployment Center**. The **Settings** tab is selected, displaying options for source, publish files, and deployment slots. A notification indicates "Save code settings" is in progress.

Uploading the zip file



Microsoft Azure portal showing the **msdocs-python-webapp-quickstart-aar** Web App in the **Configuration** page. The **Stack settings** section is visible, showing Python 3.14 as the selected stack and major version. A notification indicates "Updating web app settings" is in progress.

After successfully uploading the zip file



Final Domain Display

Post Lab Questions:

1. **How does deploying a web application on Azure IaaS differ from deploying it on a Platform as a Service (PaaS) model? (Compare responsibility, control, and management overhead.)**

Deploying a web application on Azure IaaS involves using virtual machines where the user has full control over the operating system, runtime environment, and application stack. The user is responsible for installing the OS, configuring web servers, applying security patches, managing storage, networking, scaling, and maintenance. This provides maximum flexibility but also introduces significant management overhead.

In contrast, PaaS provides a managed environment where the cloud provider handles infrastructure, operating system updates, runtime installation, scaling, security patches, and maintenance. Developers only focus on deploying application code and managing data. PaaS reduces operational complexity, speeds up deployment, and supports automatic scaling, but offers less control over the underlying system compared to IaaS.

2. **Why is Infrastructure as a Service (IaaS) suitable for deploying web applications that require full control over the operating system? (Consider OS selection, software installation, and configuration control.)**
IaaS is suitable for applications that need complete control because it provides direct access to virtual machines with customizable operating systems and configurations. Users can select the OS version, install custom software, configure security settings, modify system-level parameters, and deploy specialized middleware or legacy applications that may not be supported in managed platforms. This flexibility is essential for applications with unique dependencies, compliance requirements, or custom networking setups. Therefore, IaaS is preferred when fine-grained control over the entire computing environment is required.

Conclusion:

In this experiment, a Python web application was successfully deployed on Microsoft Azure using a Platform as a Service (PaaS) model. The application was first prepared locally, dependencies were installed, and the server was tested on localhost. A web app service was then created on Azure, and the application code was deployed using ZIP deployment. The deployed application was accessed through a public domain using a web browser, demonstrating cloud-based hosting and accessibility. This experiment illustrated how PaaS simplifies application deployment by abstracting infrastructure management while providing scalability, reliability, and ease of maintenance.